PRINCIPAL RESEARCHER



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Principal Researcher leading applied research in trusted ubiquitous physical AI, multimodal sensing, wearable and ambient computing, and responsible AI systems. I enjoy connecting the dots across projects and people to work towards broader impact.

- Research on Sensing Systems: Remote sensing, wearables and mobile, implantable, city-scale and AR. Multiple patents, open-sourced projects and publications.
- Devices experience: Device design, deployment, user testing and data analytics.
- Leadership experience: Project Inception, establish KPIs, architecture development, deployment, evaluation and enable product-partner-research conversation.
- Lead cross-organization collaborations and mentor researchers across sensing, systems, security, HCI, machine learning, hardware, and robotics.
- Research Mentorship: I have mentored over 25 graduate students and interns, many of whom are now academic faculty or applied researchers.

EXPERIENCE

2019 – present

MICROSOFT RESEARCH, REDMOND, WA

Principal Researcher (2024 – present)

- **Applied research** at the intersection of multimodal sensing, AI devices, and product incubation.
- Define and lead the vision and research efforts on **responsible ubiquitous physical AI** with a focus on user and experience-driven system development. Connecting user identity to scoped memory, environmental trust, and embodied safety across devices, shared spaces, and robots.
- Lead the cross-org efforts to architect and develop privacy-preserving user Identity and multi-modal low-power sensing for user context recognition on wearable devices as a part of Microsoft Project Solara.
- Develop privacy-aware wearable-agent systems combining selective sensing and multimodal identity verification; documented privacy evaluations reduced camera use by about 64% with about 3% lower task accuracy in the evaluated settings. Submission under review.
- Co-developed an independent edge-based robot-guardrail layer that predicts trajectory collisions and stops unsafe robot motion; to be presented as a part of ACM MobiCom 2026.
- Created scoped personal and shared memory for collaborative AI assistants, producing a working prototype and asker-dependent benchmark. To appear as a part of EMNLP 2026.
- Filing **7 patents** focused on Trusted Ubiquitous Physical AI.

Senior Researcher (2019 – 2024):

Project FoodVibes & FarmVibes — Sensing + Traceability for Global Supply Chains (2020 – 2024)

- Led applied research on ***agri-food sensing systems*** and **end-to-end traceability solutions**.
- Delivered **6 peer-reviewed publications (including in Nature Communications)**, **5 open-source GitHub projects**, and **10 patents**.
- Directed multiple research thrusts by mentoring PhD students, two MS in Technology and Innovation teams, seven cross-disciplinary interns and two visiting faculty researchers.
- Executed field deployments with Nelson Farms and **Loftus Farms,** in east Washington**,** to validate sensing and traceability systems.

- Drove Microsoft's **traceability technology strategy** for multinational food supply-chain partners in collaboration with the **Dynamics 365 Supply Chain Management team** to support **EUDR** and **FSMA** regulatory compliance.
- Represented Microsoft in **UN-affiliated working groups** shaping global policy for tech-enabled regulatory compliance in Agri-Food.
- Delivered keynotes and talks at leading universities and conference workshops to inspire the next generation of researchers

Project Eclipse — Urban Innovation Initiative (2019 – 2022)

- Co-developed and deployed a **hyperlocal air-quality sensing network** across Chicago.
- Partnered with **JCDecaux** and local community groups to instrument **100+ transit shelters** with air-quality sensors.
- Drove citizen science by spotlighting data-driven environmental monitoring.
- This data and analytics are now hosted as a part of Microsoft Planetary Computers.

Project RF Bandaid — Battery-Free & Low-Power Sensing (2019 – Present)

- Created the **world's first analog Radio Frequency Sensing Platform**, the *RF Bandaid.*
- Delivered a **technology vision review** on low-power and battery-free sensing to **Microsoft Leadership and the CEO**.
- Early establisher of a **globally recognized research direction**, enabling multiple follow-on academic collaborations. Supported students across the world with prototypes and design.
- Produced **6 peer-reviewed publications**, **5 patents**, and **open-sourced hardware and system designs** to academic researchers worldwide.

2022 – present **UNIVERSITY OF WASHINGTON, ECE**
Affiliate Asst. Professor

- I mentor undergraduate and graduate students conducting research at the UW.
- I am also a launch project mentor for MSTI Student teams from the UW Global Innovation Exchange (GIX) program.
- Served on the graduate committee for PhD students at top tier universities worldwide, including University of Washington, CMU, UIUC, University of Wisconsin and Upsala.

EDUCATION

2013 – 2018 **UNIVERSITY OF WASHINGTON, SEATTLE, WA**
PhD. In Electrical Engineering
Dissertation: Wireless Biomedical Sensing

2011 – 2013 **CASE WESTERN RESERVE UNIVERSITY, CLEVELAND, OH**
MS. in Electrical Engineering and Computer Science
Thesis: Silicon Carbide NEMS Logic and Memory for Computation at Extreme: Device Design and Analysis

2007 – 2011 **AMRITA UNIVERSITY, COIMBATORE, INDIA**
B.Tech. in Electronics and Instrumentation Engineering

RESEARCH SUMMARY

Publications and Patents: Google Scholar, Google Patents

Microsoft Research: (Redmond,WA)

- **Project Food Vibes & FarmVibes**: Traceability and Sensing for Global Food Supply Chain
  - Led a team of researchers, developers and interns towards technological innovations that can enable a transparent and sustainable food supply chain. Our team identified technical gaps and addressed them through innovation in remote, hyperlocal and wireless sensing.

- Worked with customers & partners to establish the KPIs and address gaps in food supply chain traceability and management. This includes tracking produce post-harvest along with quality, sustainability and other geospatial metrics while meeting global regulatory compliance.
- Represented Microsoft at UN and ISO affiliated standards workgroups: White papers
  - Forest Monitoring and Remote Sensing Guidelines: Whitepaper link
  - Farmer Income and Production Cost Tracking: Whitepaper link

- **Project Eclipse: Hyperlocal Air Quality Sensing:**
  - Enabled Hyperlocal and low-cost air quality sensing for cities. Support equitable environmental sensing through a publicly accessible city-wide deployment.
  - Enabled citizen science that identifies Chicago's air pollution hotspots. Media link
  - The Project Eclipse team operated with the aim to spotlight environmental sensing, esp. urban air quality, and empower local communities with knowledge-driven decision making.

Graduate PhD. Research: (University of Washington, Seattle)

- **Brain-Computer-Spinal-Interface (BCSI): A Fully Wireless Neural Interface**
  - Research to restore motor function in patients with paralysis caused by spinal cord injury. I developed a closed-loop BCSI implant to enable neural recording and stimulation. This device featured wireless energy harvesting and low-power communication & computation.
  - The fully wireless, miniature printed circuit device with neural recording/stimulation capability has been used by neural engineering collaborators for rehabilitation study in mice.
  - The work featured an approximate computing core, Wireless Power Transfer and low-power Backscatter Communication to make it wire-free and fully implantable.
  - This work was awarded the Washington Research Foundation fellowship and was the Madrona Prize runner up in 2018.
- **RF Bandaid: Fully-Analog Battery-Free Wireless Interface for Wearable Sensors**
  - Developed an **ultra-low-power wearable sensor platform** operating at **40–160 µW**, powered entirely via RF energy harvesting. An open design used by many researchers worldwide.
  - Built **modular, plug-and-play platform** demonstrated for **heart rate, breathing rate, temperature, and audio/sound sensing**.
  - I continue to support research in low-power & analog sensing.

Graduate MS. Research: (Case Western Reserve University, Cleveland)

  - **Nano Electromechanical Switches for Extreme Condition Operation**
  - Designed and fabricated low power SiC NEMS switches for high temperature (>600°C) logic.
- **Miniaturized Ultrasonic Device for Detection and Monitoring of Cancer Recurrence**
  - Co-developed and tested the electronics for an implantable ultrasonic assembly for early detection and monitoring of cancer and subsequent treatment planning [Article].
- **Undergraduate Research: (Indian Institute of Technology, Bombay)**
  - **Bio-MEMS based Instrumentation for Glucose and Urea Sensing**
  - I implemented optical sensing circuits and micro-electromechanical systems for enzyme-based glucose and urea sensors.

PROFESSIONAL LEADERSHIP AND SERVICE

- **Conference Organization & Program Committee:** I continue to serve on the technical program committee for top ACM and IEEE conferences in mobile and sensing systems (like MobiSys, CHI, RFID and Sensys) and have served as a project reviewer the NSF RINGS program.
- **Journal:** Reviewer for Nature, Sensors and Transactions. **Associate Editor**: IMWUT Journal 2019-current.
- I am also a strong proponent of **STEM and Women in Engineering**. I believe that education is one of the best ways to empower a person. Towards this, I frequently volunteer with early education non-profits (like GoLaadli) and academic groups, through programs like WAFER at the University of Washington.
- I continue to serve as a mentor to women in STEM undergraduate and graduate programs worldwide.